Problem 1

Task: Design a linear state-feedback controller for the plant $G(s)$ that gives 5% overshoot and peak time of 0.3 second.

$$G(s) = \frac{100(s+10)}{s(s+3)(s+12)} \tag{1}$$

Design steps:

a) Find the characteristics equation (hint: add a pole at -10 to cancel the zero)

b) Find the feedback gain K

Problem 2

Task: Consider the DC motor system shown in Figure 1 that is described by the equations in 2.

$$\begin{aligned} L\frac{di}{dt} + Ri &= u - \Psi\omega \\ J\frac{d\omega}{dt} + B\omega &= \Psi i \end{aligned} \tag{2}$$

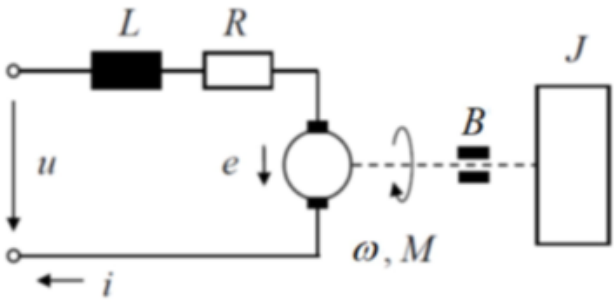


Figure 1: DC motor

Parameter	Value
R	1
L	0.0002
Ψ	0.04
B	0.0001
J	0.00005

1. Convert the system into state-space form.
2. Evaluate the controllability of the system.
3. Design full state-feedback controller with closed-loop system poles twice as faster than open-loop poles (real-part).
4. Design a pre-filter that ensures the accurate tracking.

Problem 3

Task: Design a state-feedback controller for the plant $G(s)$ such that closed loop poles have $\zeta = 0.7$ and $\omega_n = 2$.

$$G(s) = \frac{1}{s(s+10)} \tag{3}$$

System states are not measured.

1. Write the system in observer canonical form.
2. Find the closed loop characteristics equation.
3. Design a full state-observer such that eigen values of $(A - LC)$ are $-10, -10$.

Problem 4

Task: Consider the system.

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u \tag{4}$$

Design a state-feedback controller such that closed loops poles are at $s = -2 \pm j4, s = -10$.